

Performance Comparison of Machine Learning Models for Image Classification

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Introduction :

The key focus of this project is on the construction steps of CNN architectures and discussion of their accuracy, as well as analyzing the differences between CNNs and traditional machine learning models in image classification.

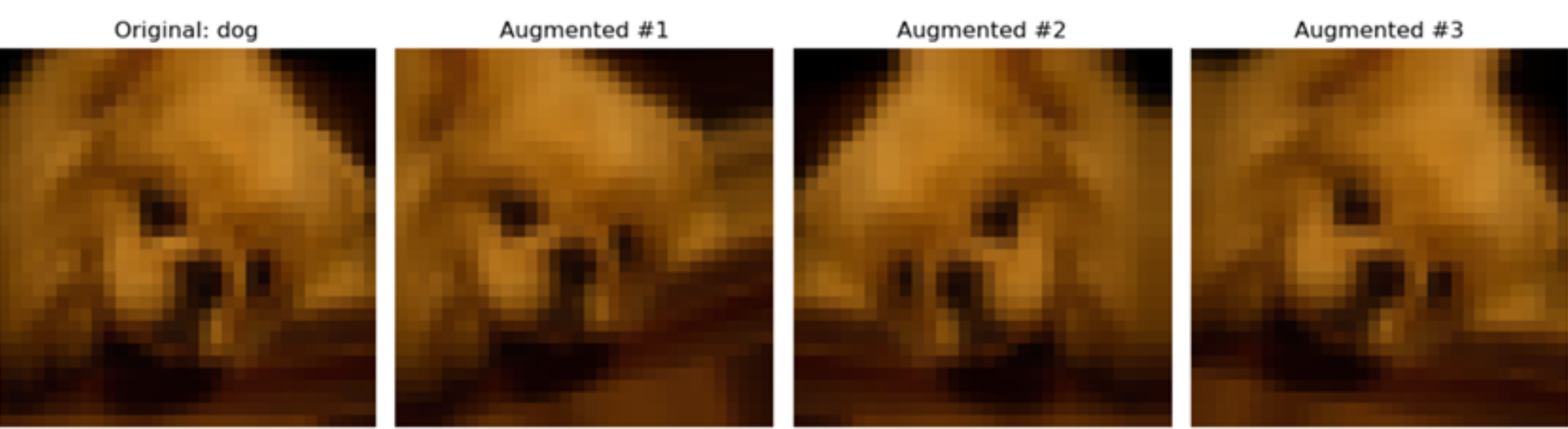
Dataset:

CIFAR-10 Dataset: Consists of 50,000 color images of size 32×32 pixels, categorized into 10 classes (such as airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck). Among them, 40,000 images are used for training and 10,000 for testing, with a balanced number of samples in each class.

Deep learning Method

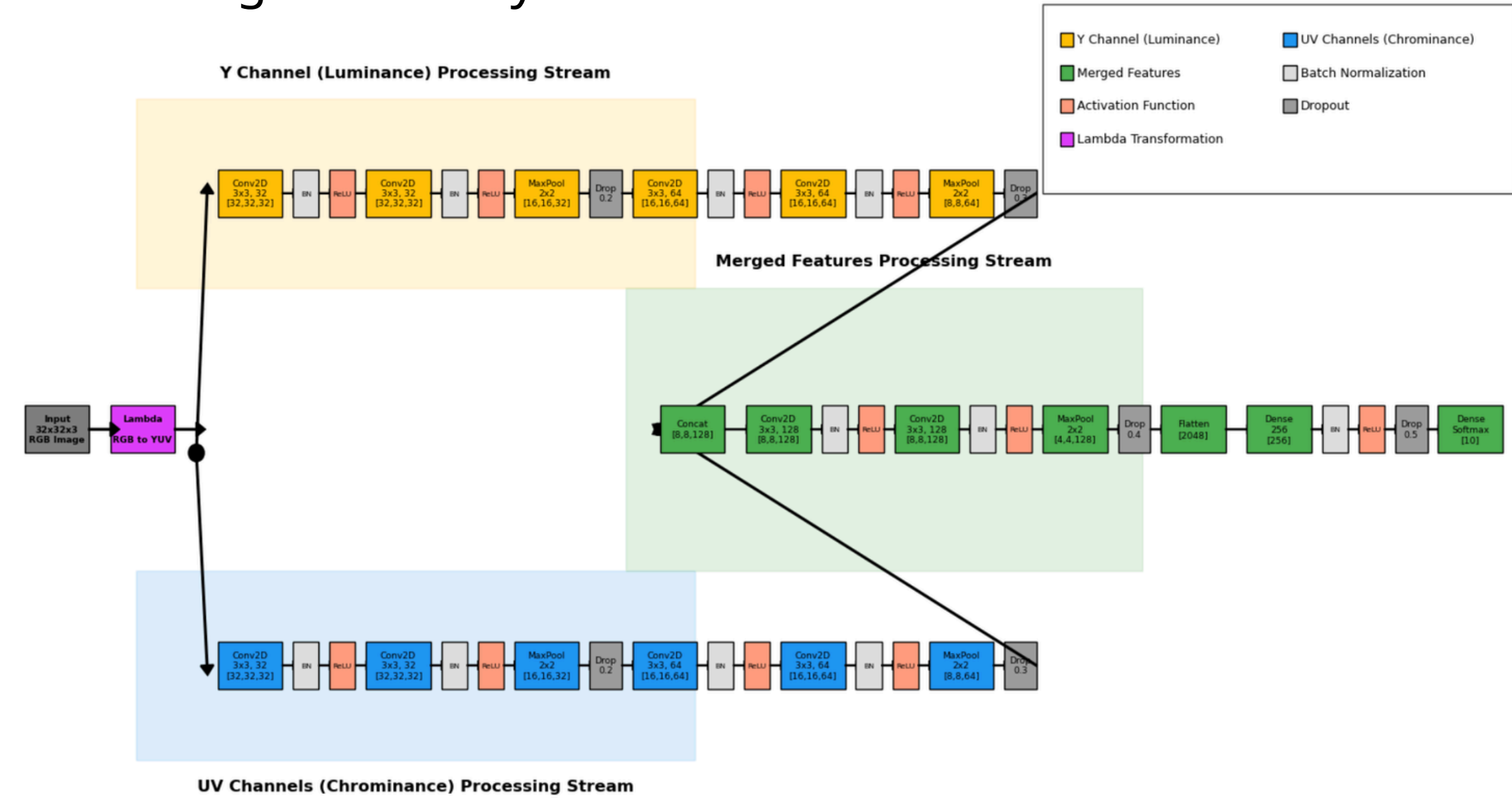
Step 1: Data Augmentation

- Random rotation up to ±15 degrees
- Random horizontal and vertical shifts up to 10%
- Random horizontal flips
- Random zoom up to 10%
- This process increases our training data from 40,000 to 160,000 samples.



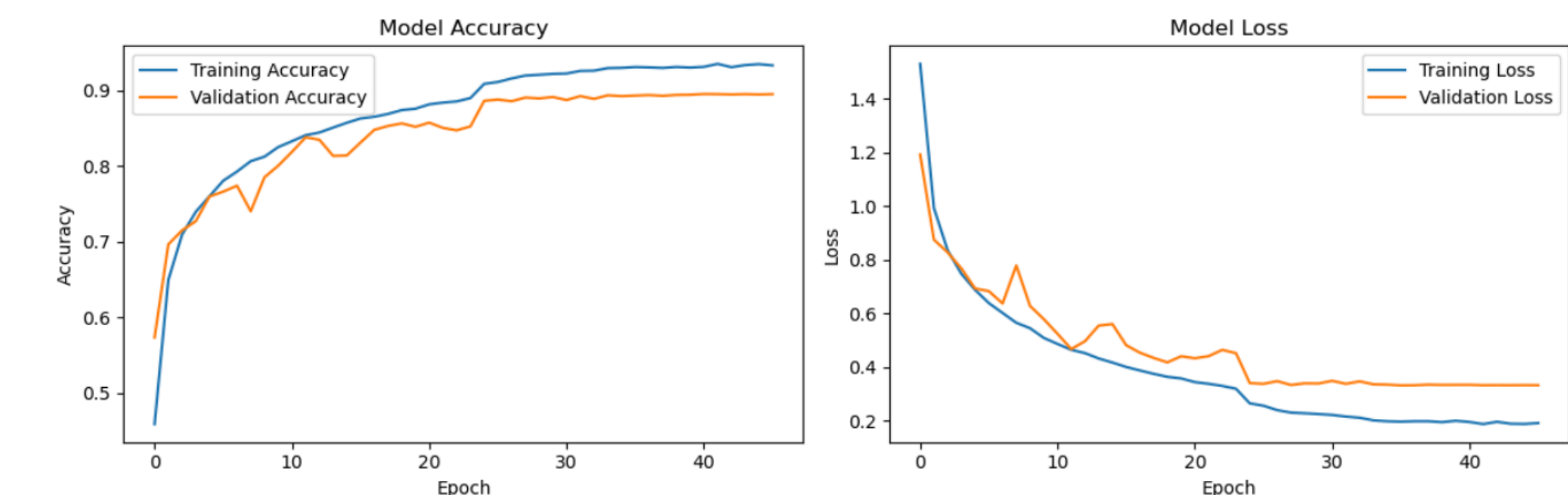
Step 2: Build the Model

- Converts RGB images to YUV color space, separating brightness and color information
- Processes luminance (Y) through one dedicated stream for brightness features
- Processes chrominance (UV) through a parallel stream for color features
- Both streams use convolutional layers, batch normalization, ReLU activations, and dropout
- Features from both streams are concatenated after independent processing
- Combined features undergo additional convolutional processing before flattening
- Final classification into 10 categories is performed through dense layers with softmax activation



Step3 : Result

The CNN model demonstrates excellent performance with training accuracy reaching up to 93% and test accuracy achieving 90%.



Traditional machine learning Method

Trained three models on the full data:

All models underwent hyperparameter optimization using GridSearchCV with 3-fold cross-validation to prevent overfitting.

Initial Classification Model Performance

Model	Cross-Validation Accuracy	Training Time (s)
SVC	0.4350	3251.12
Random Forest	0.4098	954.01
Logistic Regression	0.3396	1348.55

PCA Dimensionality Reduction Classification

Results showed that SVC performed best in cross-validation (64.04%). PCA not only maintained model performance but also significantly reduced computational requirements.

PCA + Classification Model Performance

Model	PCA Component	Cross-Validation Accuracy	Test Accuracy	Training Time (s)
PCA + SVC	500	0.6404	0.5763	3217.29
PCA + Random Forest	100	0.4656	0.4380	967.5
PCA + Logistic Reg	100	0.4180	0.3826	943.82

Conclusion

Performance gap:

CNN achieved 93% training and 90% test accuracy, significantly outperforming traditional methods where even the best model (SVC with PCA) reached only 64.04% accuracy

Architectural differences:

- CNNs are generally considered "black box" models compared to some traditional ML models
- The complex architecture and large number of parameters make CNNs harder to interpret
- Traditional models like decision trees or linear models often have more transparent decision-making processes